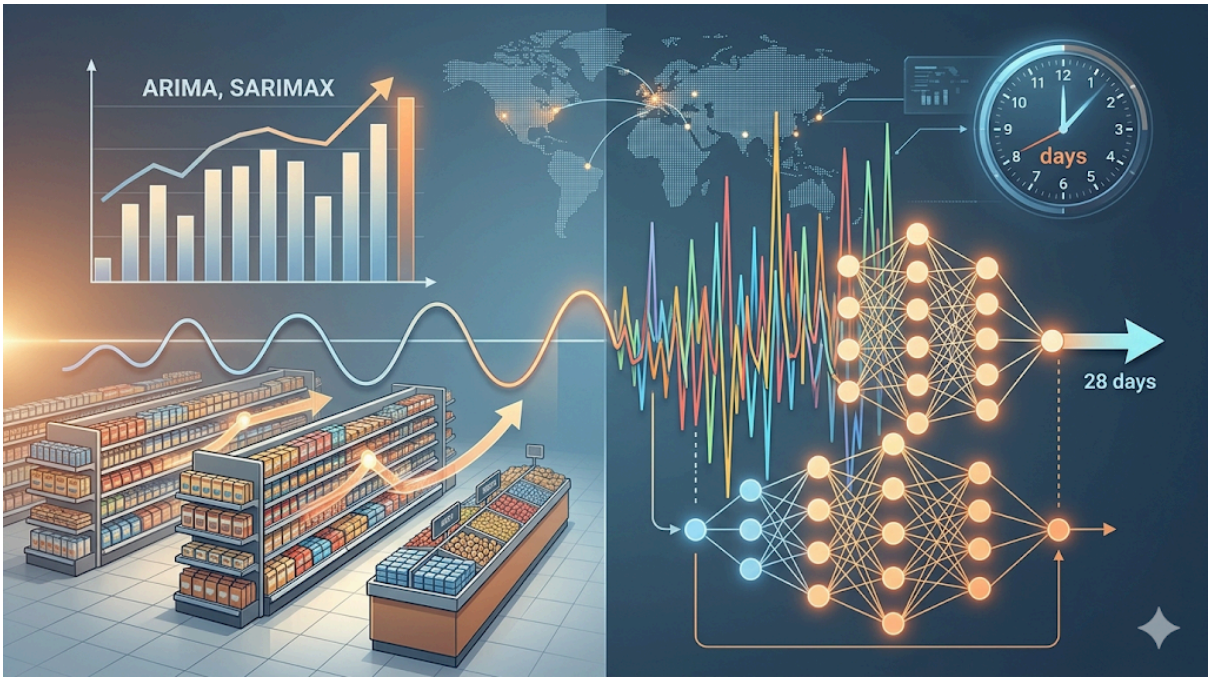
Advanced Time Series Forecasting in Retail: A Comparative Analysis from Classical Statistics to Deep Learning

Andoni Garrido, Pablo Ordoyo y Mattin Arluciaga

GitHub repository: *https://github.com/mattarluu/AML-Project*



Universidad de Deusto
Deustuko Unibertsitatea
University of Deusto

INDEX

The Selected Topic **3**

Academic Justification and Syllabus Alignment 4

Approach and Main Objective 4

Selected Dataset and Exploratory Data Analysis (EDA) **5**

Structural Components: Trend and Seasonality 5

Exogenous Variables and Economic Context 7

Micro-Level Dynamics and Cross-Item Correlation 7

Statistical Foundation for Classical Models 9

Planning of Project's Content **10**

Phase 1: Validation Strategy and Data Leakage Prevention 10

Phase 2: Algorithmic Baselines 10

Phase 3: Univariate Statistical Modeling 10

Phase 4: Multivariate Statistical Modeling 11

Phase 5: Deep Learning and State-of-the-Art Approaches 11

Phase 6: The Model Tournament and Final Conclusions 11

The Selected Topic

The core theme of this project is time series forecasting applied to retail demand, an environment characterized by high mathematical complexity. Unlike continuous and stable time series (such as temperature or high-volume financial markets), daily product sales in supermarkets present a significantly greater challenge: intermittent demand (an abundance of zero values), strong seasonality (weekly micro-cycles and annual macro-cycles), and high sensitivity to exogenous variables (price fluctuations, special events, and macroeconomic subsidies).

The problem to solve consists of accurately predicting the daily sales volume at the item level over a 28-day forecasting horizon, minimizing the error to optimize supply chain operations.

Academic Justification and Syllabus Alignment

This project has been designed to integrate and practically apply the key concepts of the Advanced Machine Learning course:

- **Methodology and Data Leakage Prevention:** Given the sequential nature of the data, the project will apply a strict temporal validation methodology. A chronological split (Train/Validation/Test) will be implemented to ensure the total isolation of the target variable and future covariates, guaranteeing a robust evaluation free of information leakage.
- **Evolution of Statistical Modeling:** A baseline error limit will be established using Naïve models. Subsequently, the project will advance to univariate statistics (ARIMA) to mathematically model autocorrelation, followed by a leap to multivariate models (SARIMAX) to evaluate the empirical impact of injecting exogenous variables (calendar events and pricing) into the mathematical equation.
- **State of the Art in Forecasting:** To address the intermittency problem and the limitations of local models, the final phase of the project will scale up to Neural Networks (Deep Learning, such as LSTMs) and foundational LLM-based models (such as Chronos). The objective is to exploit the capacity of these algorithms for cross-learning (learning global patterns by processing multiple time series simultaneously).

Approach and Main Objective

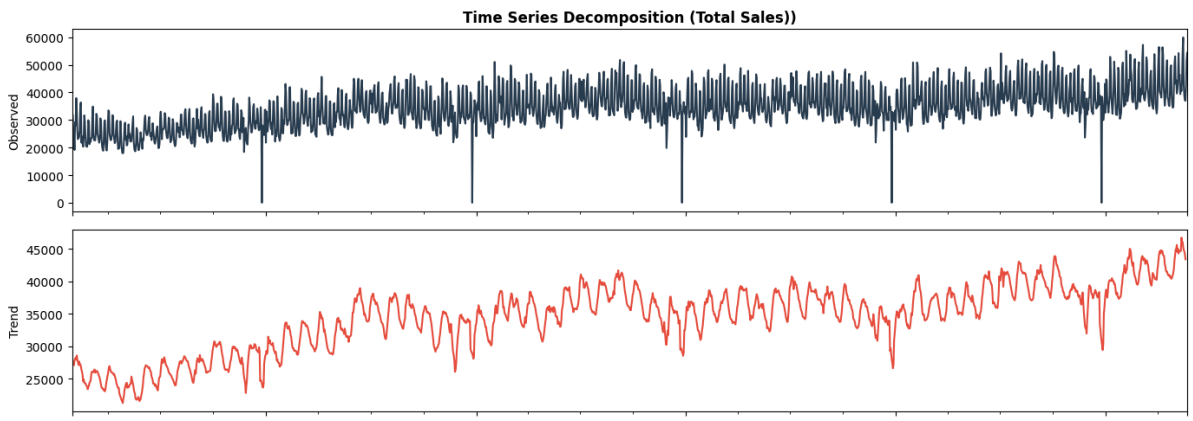
The methodological core of the project will be the execution of a **"Model Tournament"**. Instead of seeking a single winning algorithm from the start, a funnel of ascending complexity will be built. All models will be evaluated against the exact same target: predicting the sales of the Top 10 highest-turnover products in a specific store. The ultimate goal is to document and quantify the trade-off between the gain in predictive accuracy (RMSE/MAE reduction) and the increase in computational cost as the project transitions from classical statistics to artificial intelligence.

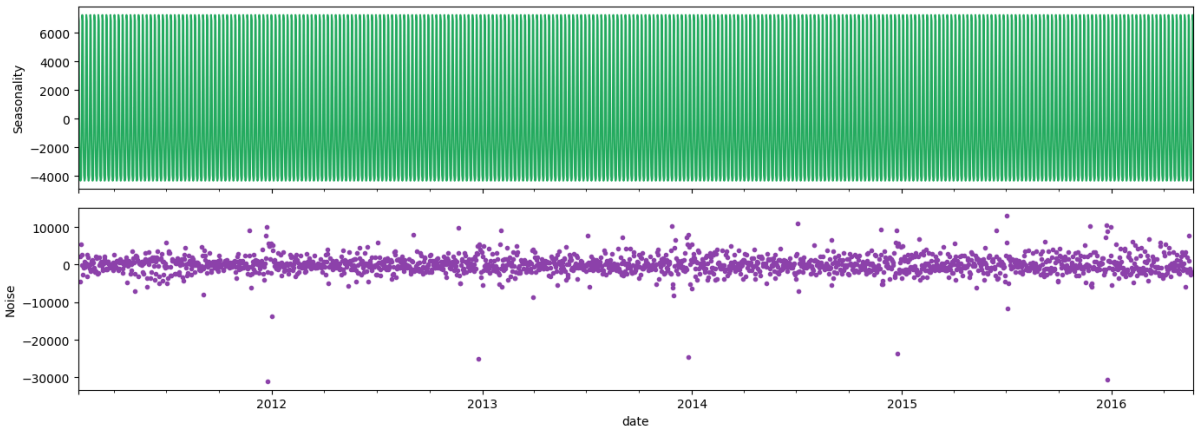
Selected Dataset and Exploratory Data Analysis (EDA)

Dataset: https://www.kaggle.com/datasets/aryayadav0513/m5-forecasting-accuracy

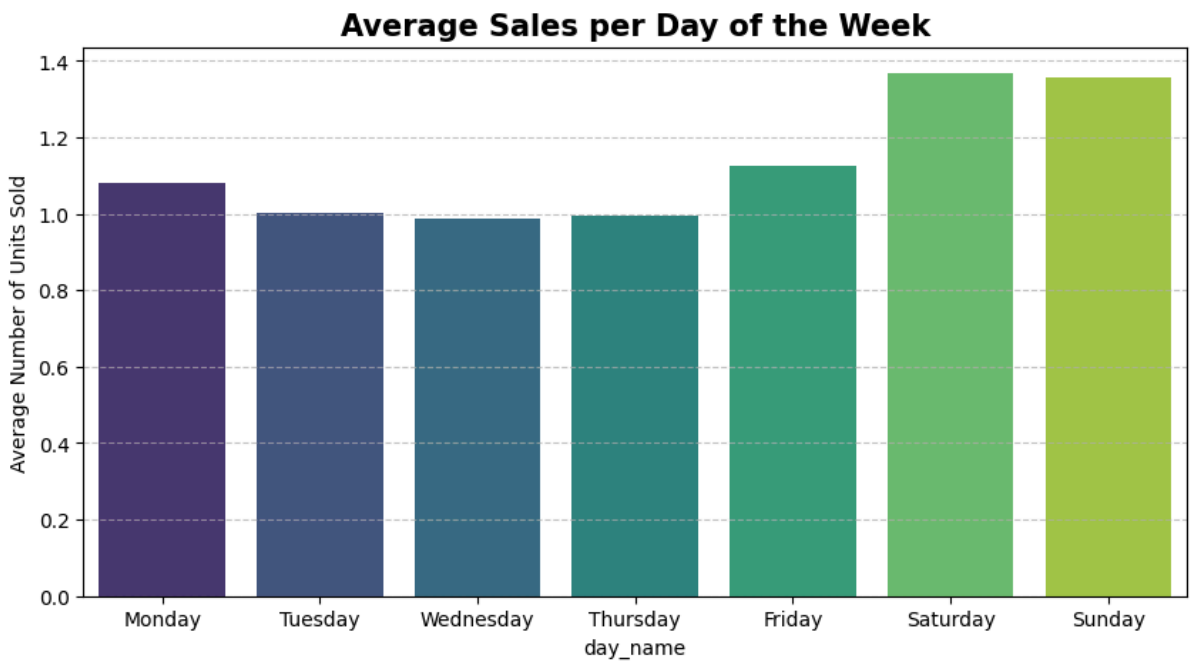
The dataset selected for this project is the **M5 Forecasting - Accuracy**, provided by Walmart. It contains hierarchical sales data, calendar events, and historical pricing. To prepare the data for sequential algorithms, a substantial data wrangling process was executed. The original dataset, structured in a "wide" format (days as columns), was melted into a "long" format (days as rows). To handle the computational load of merging millions of rows, a downcasting technique was applied, optimizing numerical data types. This memory optimization allowed for the seamless merging of the sales history with the calendar and pricing tables without causing hardware bottlenecks.

Structural Components: Trend and Seasonality



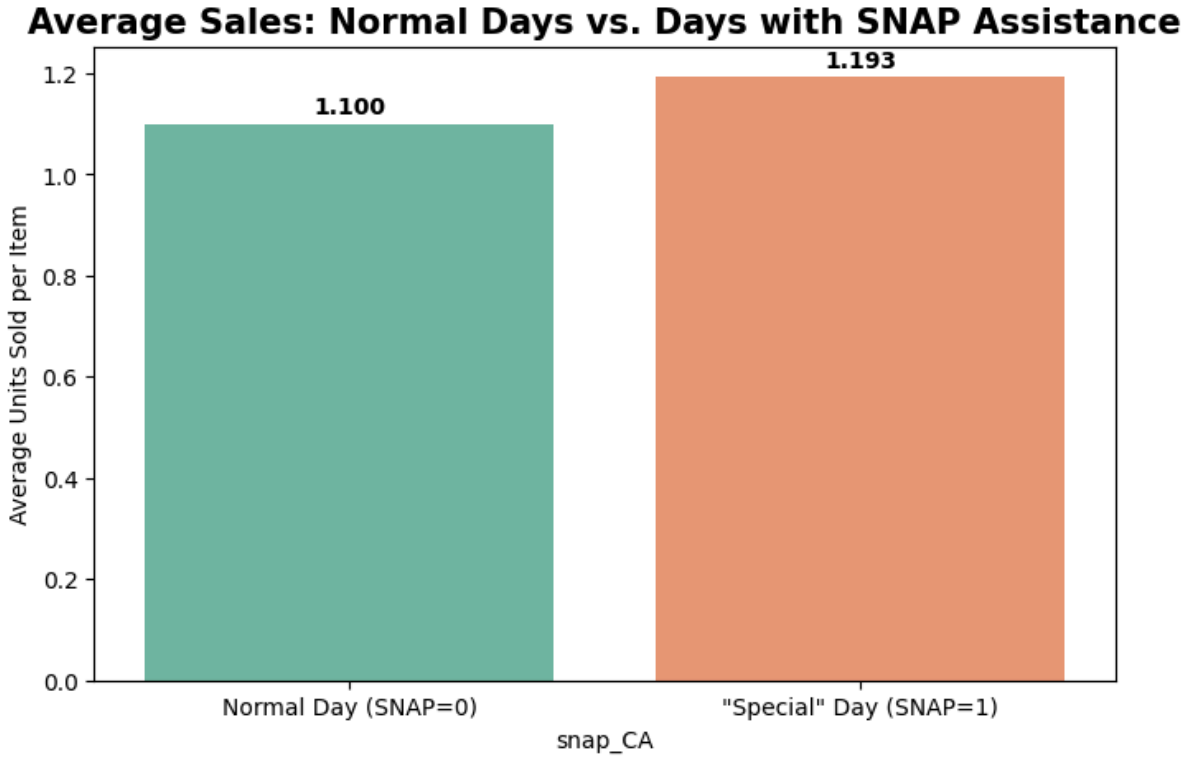


Applying a seasonal decomposition algorithm to the aggregated series isolates its fundamental mathematical components. The "Trend" subplot reveals a steady, positive upward trajectory in overall sales from 2011 to 2016. Furthermore, the "Seasonality" component displays a highly dense and perfectly regular oscillating pattern. The "Residuals" (noise) highlight periods of high volatility and sudden drops (such as the annual Christmas closures where sales plummet to zero), confirming the stochastic nature of retail demand.



Zooming into the micro-cycles, the empirical evidence shows a heavily pronounced weekly seasonality. The bar chart clearly indicates that sales volume spikes significantly during the weekends (Saturday and Sunday), while remaining relatively flat during the weekdays. Any autoregressive model deployed in the next phases must inherently account for this 7-day cyclical behavior.

Exogenous Variables and Economic Context

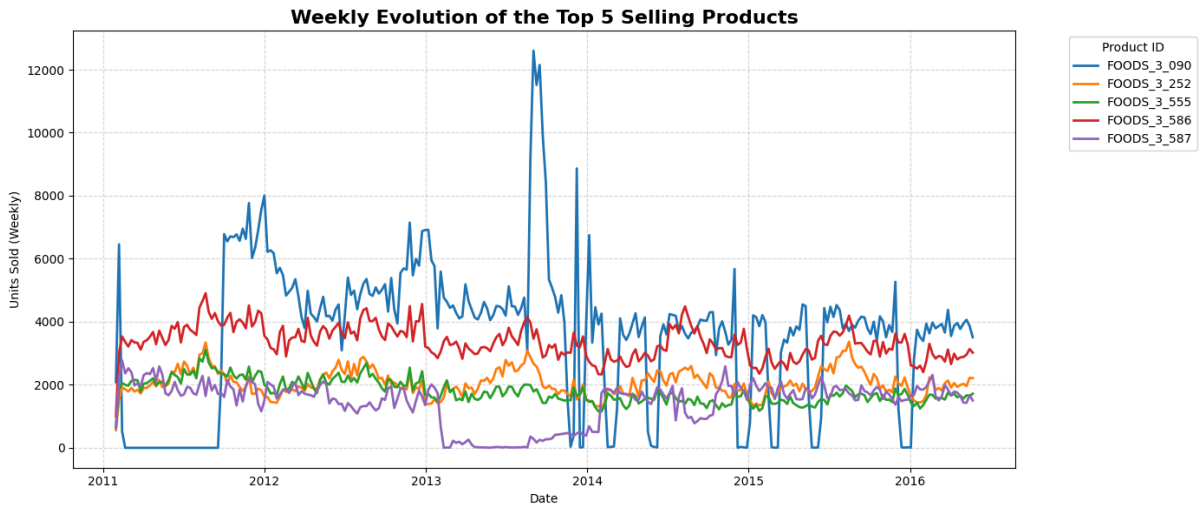


Time series in retail do not operate in a vacuum. This bar chart explores the tangible impact of exogenous variables, specifically the days when the government injects liquidity through the Supplemental Nutrition Assistance Program (SNAP).

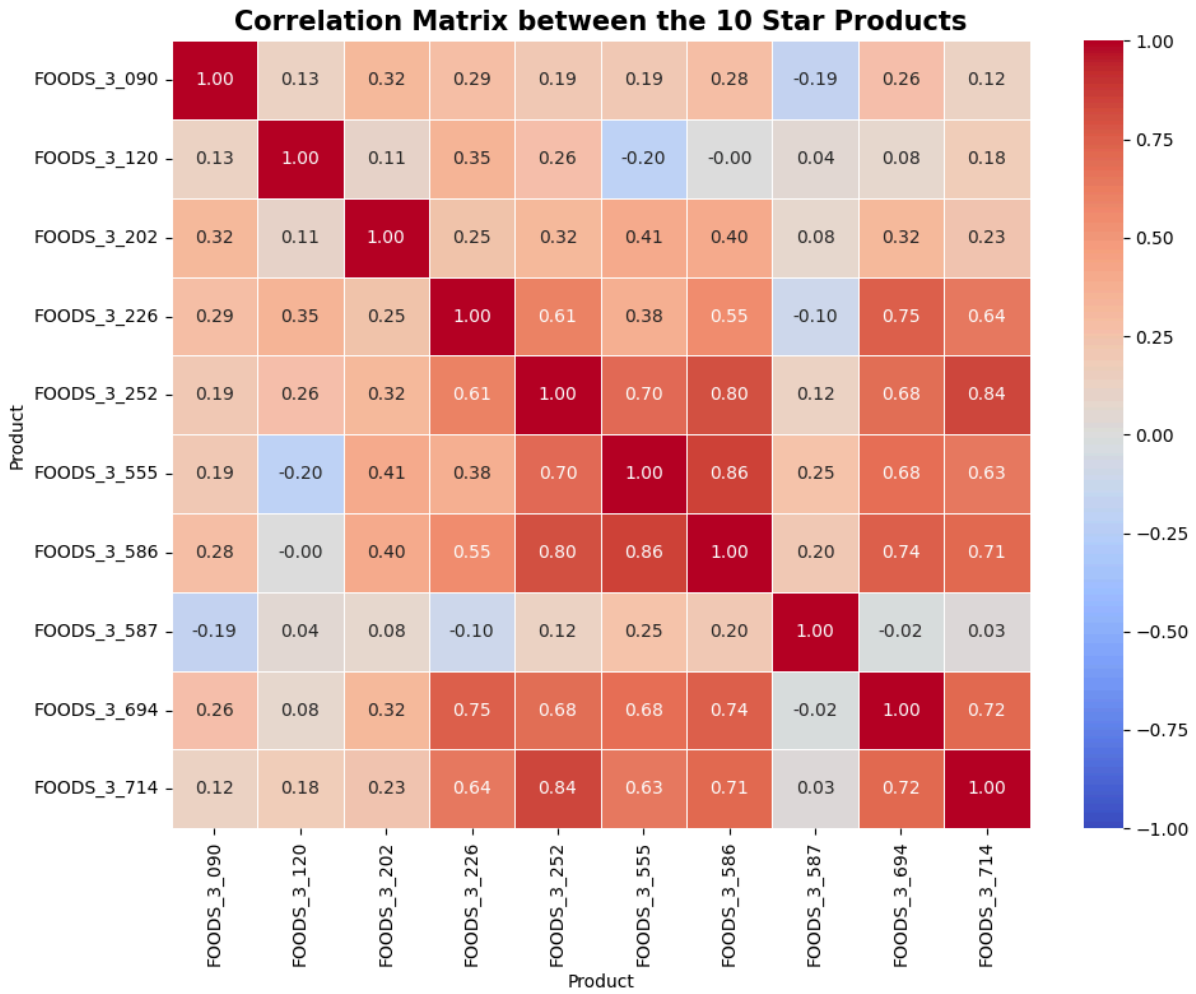
By comparing the mean sales volume, we observe a quantitative lift in average unit sales per item on days with active subsidies (1.193 units) compared to normal days (1.100 units). This empirical evidence validates the theoretical necessity of testing multivariate models (SARIMAX) to capture these external economic drivers, as they clearly alter baseline consumer purchasing behavior.

Micro-Level Dynamics and Cross-Item Correlation

To establish a level playing field for the "Model Tournament", the target scope was narrowed down to the Top 10 best-selling items in the FOODS_3 department of the CA_1 store.

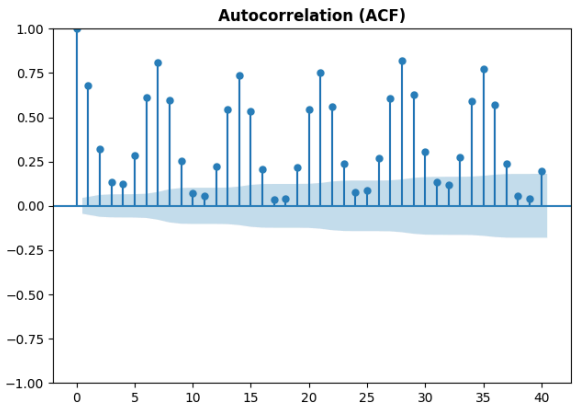


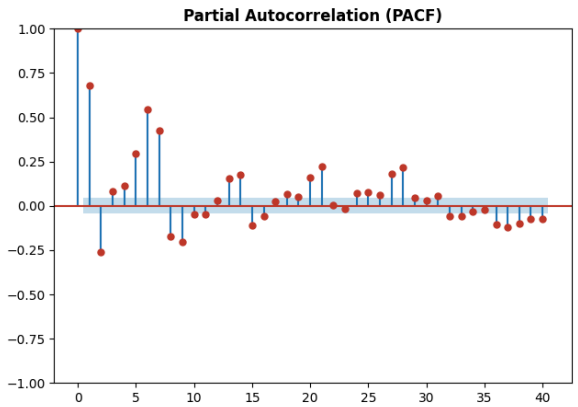
Isolating the time series at the individual item level reveals a much higher degree of variance, noise, and intermittency compared to the aggregated macro view. Forecasting these chaotic individual lines, characterized by sudden peaks and periods of zero demand, is the true mathematical challenge of the project.



The correlation matrix of the Top 10 items is a foundational piece of evidence for this project. The heatmap reveals significant positive linear relationships between several top-selling items (strong correlations exceeding 0.80). This mathematically proves that these products share underlying purchasing behaviors. Classical univariate models are "blind" to this cross-item synergy, providing the ultimate academic justification for deploying Deep Learning architectures (such as LSTMs or Chronos) in the final phase, which can leverage cross-learning by processing all series simultaneously.

Statistical Foundation for Classical Models





Before configuring the classical statistical baseline models, an analysis of stationarity and autocorrelation was performed. The Autocorrelation Function (ACF) plot visually confirms the seasonal findings, displaying highly significant spikes exactly every 7 lags (7, 14, 21, 28, etc.). The Partial Autocorrelation Function (PACF) shows a strong immediate dependency on the most recent prior days. These graphical insights will be strictly utilized to define the initial autoregressive and moving average parameters for the forthcoming ARIMA and SARIMA implementations.

Planning of Project's Content

To ensure a rigorous, coherent, and methodologically sound development of the "Model Tournament", the project will be structured into six consecutive phases. This phased design provides a clear analytical progression, whereby each increase in model complexity is motivated by the empirical limitations identified in the preceding approaches. In this way, the project does not simply compare a collection of algorithms, but rather follows a systematic path from simpler baseline methods to more advanced forecasting techniques.

Phase 1: Validation Strategy and Data Leakage Prevention

To establish a reliable evaluation framework that preserves the temporal structure of the dataset, a strict chronological split will be implemented. The data will be partitioned into three non-overlapping subsets: a Training set (Days 1–1885), a Validation set (Days 1886–1913), and a Test set (Days 1914–1941).

By completely isolating the final 28-day test horizon throughout the model development stages, this approach replicates a realistic forecasting scenario and ensures an unbiased estimate of predictive performance during the final evaluation, effectively preventing any data leakage.

Phase 2: Algorithmic Baselines

This phase aims to establish a set of reference benchmarks that define the minimum acceptable level of predictive performance. To achieve this, zero-intelligence forecasting models will be implemented and assessed: the Naïve approach, which predicts the previous day's value, and the Seasonal Naïve approach, which predicts the value recorded exactly one week earlier.

Their performance will be measured using RMSE and MAE, thereby establishing the baseline error thresholds that more advanced models must meaningfully outperform in order to justify their additional complexity.

Phase 3: Univariate Statistical Modeling

Focusing on the internal temporal dynamics of each series, this stage captures linear trends, seasonality, and autocorrelation patterns without incorporating

exogenous variables. Classical univariate models, specifically Exponential Smoothing (Holt-Winters) and SARIMA methodologies, will be developed for each of the Top 10 items.

The autoregressive, differencing, and moving average parameters (p, d, q) will be carefully calibrated based on the Autocorrelation (ACF) and Partial Autocorrelation (PACF) analyses conducted during the exploratory data analysis.

Phase 4: Multivariate Statistical Modeling

To evaluate the extent to which forecasting performance can be improved by incorporating relevant contextual information, the univariate framework will be extended into a SARIMAX architecture. This phase will inject exogenous variables identified as significant in the EDA—such as SNAP subsidies, calendar events, and historical price variations—into the mathematical equation. This allows for a direct assessment of the reduction in forecasting error achieved when the model can account for anomalous demand periods using external covariates.

Phase 5: Deep Learning and State-of-the-Art Approaches

Addressing the limitations of local, linear models, this phase leverages global learning frameworks capable of capturing non-linear dynamics and cross-series dependencies. Advanced architectures, such as a Long Short-Term Memory (LSTM) network or foundational Time Series LLMs (e.g., Chronos), will be designed and trained.

The strategy will shift from item-specific training to global modeling, processing the entire dataset simultaneously to exploit shared temporal structures and inter-item correlations through cross-learning.

Phase 6: The Model Tournament and Final Conclusions

In the final phase, all fully trained models will be evaluated on the untouched 28-day Test set to integrate the results and identify the optimal forecasting approach. A comprehensive framework will be compiled to contrast forecasting accuracy (RMSE, MAE) against computational cost and practical scalability.

Ultimately, the final report will conclude whether the transition from classical multivariate models to Deep Learning yields a justifiable, statistically significant advantage in a retail forecasting context.